SmartTask : Context-Aware Mobile To-Do List for Intelligent Task Management

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SmartTask

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Line-Up

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invideo/smarttask

Problem & Impact

- No Context Awareness
- Static task prioritization
- Poorly timed notifications

Most existing to-do

- frequent
- missed pr
- increased
- reducing

How the Solution Addresses the Problem?

Problem

Solution

Most to-do apps ignore user context, causing poorly timed reminders, interruptions, and extra mental load.

Solution :

- **Smart Prioritization →** Tasks are automatically prioritized using real-time user context
- **Smart Notification Timing →** Notifications are delivered only when the user is receptive
- **Focus-Aware Alerts →** Alerts are suppressed during meetings, travel, or focus time
- **Reduced Cognitive Load →** System shows only the most relevant tasks

Solution Overview

SmartTask is a context-aware mobile to-do app that prioritizes tasks and sends reminders based on the user's real-time context such as location, activity, and schedule.

Key Features :

- **Context-Aware Task Prioritization** → Tasks are automatically ranked using user context
- **Adaptive Notification System**→ Reminders are sent at the most appropriate moments to avoid interruptions.
- **Behavioral Learning & Insights** → The system learns from user actions to improve future task suggestions.

Solution Overview

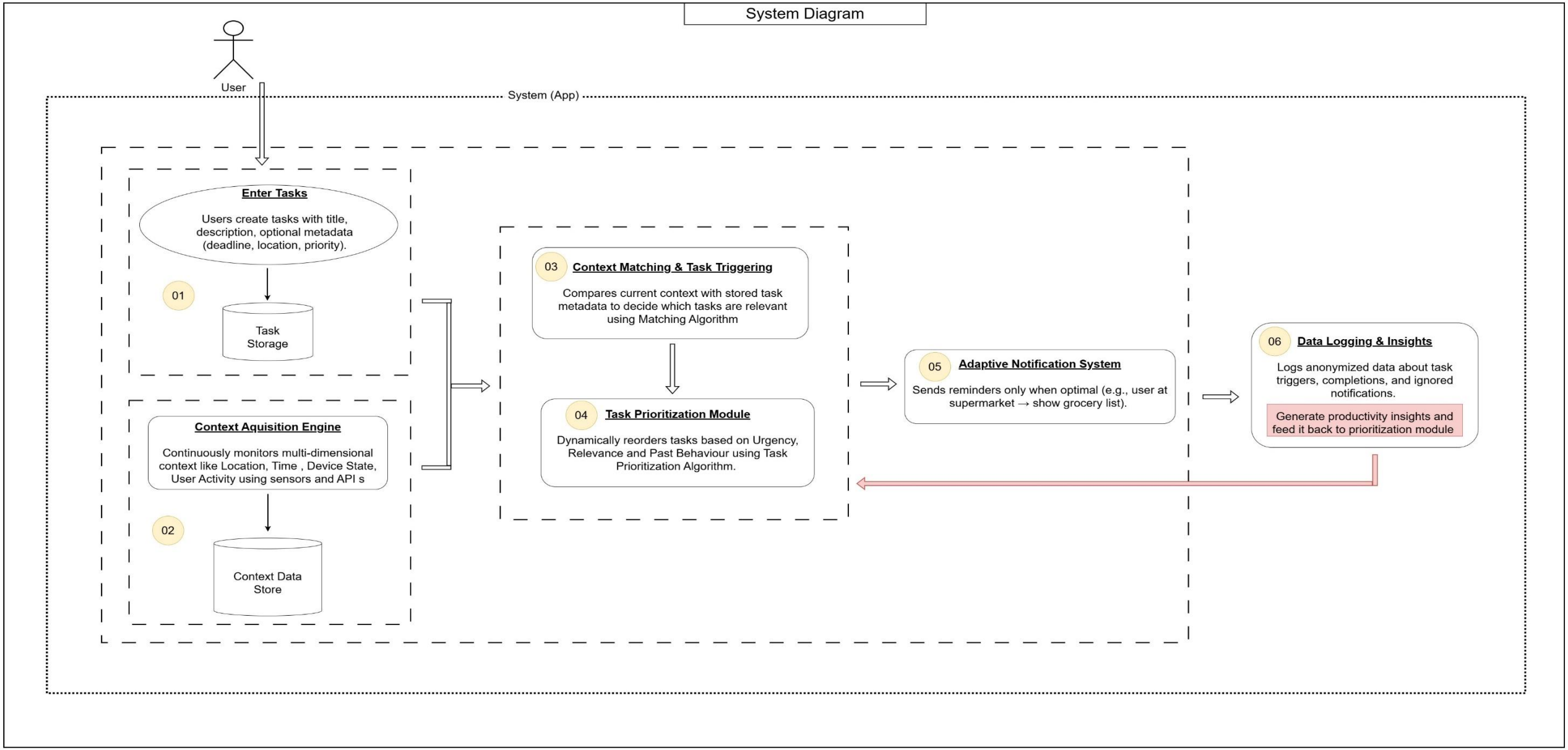
Non-Functional Focus :

- **Performance** → Real-time task prioritization and quick notifications
- **Security & Privacy** → User data stored locally
- **Usability** → Simple, intuitive mobile interface
- **Reliability** → Basic features works offline

Professional, Legal, Social, Security & Ethical Issues :

- **Privacy** → Context data stored locally
- **User Consent** → Permissions required for location, activity, and calendar access
- **Security** → No personal data shared externally
- **Professional Responsibility** → Ethical data usage and transparent system design

System Design: Six - Component Architecture



System Design: Components

- **Task Input –** manual task creation and editing
- **Voice Command Task Creation –** captures spoken task commands and converts them into structured task fields. (additional component suggested by examiner panel)
- **Context Acquisition Engine –** gather current user context
- **Context Matching & Triggering –** filter tasks relevant to the current context
- **Task Prioritization–** ranks tasks based on urgency and relevance
- **Adaptive Notification System –** sends reminders at suitable times
- **Data Logging & Insights –** records user actions for future improvement

System Design : Components Flow In Real World

Scenario: A student has 3 upcoming tasks today:

1. Finish Assignment – Deadline: 2:00 PM | Location: Library | Priority: High
2. Buy Stationery – Deadline: Evening | Location: Bookshop | Priority: Medium
3. Call Family – Deadline: Night | Location: Home | Priority: Low

1 - Task Input Module

- Student enters tasks with metadata: title, deadline, location, priority.
- Example: "Finish Assignment – 2:00 PM – Library – High Priority".
- Stored in Task Database.

2 - Context Acquisition Engine

- Continuously gathers real-time signals:
  - Location → detects when student is at home or library.
  - Time → checks system clock (e.g., 1:30 PM, 30 mins before deadline).
  - Device state → battery at 20% triggers early reminder.
  - Activity → detects "walking" → student is moving toward library.
- Stored in Context Data Store.

3 - Context Matching & Task Triggering

- Matches current context with task metadata.
- Example: At 1:30 PM, location = near library, time = 30 mins to deadline.
- Filters task list → highlights "Finish Assignment" as relevant.

4 - Task Prioritization Module

- Orders tasks dynamically:
  - Finish Assignment → top priority (deadline + location match).
  - Buy Stationery → lower (not urgent now).
  - Call Family → lowest (later at night).
- Works in loop with insights to refine ranking.

5 - Adaptive Notification System

- Sends just-in-time reminder:
  - "You're near the library, time to finish your assignment before 2:00 PM."
- Suppresses irrelevant notifications (Call Family won't pop up now).

6 - Data Logging & Insights

- Logs: student opened the reminder, completed assignment at 1:55 PM.
- Stores response + context.
- Feeds back into prioritization → system learns optimal reminder timing (e.g., 30 mins before worked well).

Result:

- Student finishes assignment on time with minimal distractions.
- Notifications felt timely and useful.
- System adapts to user behavior for future days.

Commercialization Potential

Business Model – How We Make Money

- **Differentiation:** Unlike other apps, ***SmartTask*** is adaptive & context-aware, offering intelligent task prioritization and notifications tuned to real-life conditions.
- **Investment & Cost:**
  - Labor: Development hours for AI algorithms, UI/UX, backend.
  - Material: Android devices for testing.
  - API s : Subscriptions Cost for API s (e.g. google maps).
  - Recovery: Freemium approach – free version with ads; premium subscription offers advanced insights and an ad-free experience.
- **Target Market:** Students, busy professionals, productivity enthusiasts.
- **Customer Profile:** Mobile-first users with heavy task-list dependency and fast-paced lifestyles.
- **Value Proposition:** Reduce stress, improve productivity, and deliver notifications at the right moment, not just the next moment.
- **IP Considerations:** Proprietary algorithms for context-matching and adaptive notification timing create defensible intellectual property.

SmartTask : A Context-Aware Mobile Task Management System

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App Demo

Q & A ??

Thank You !!!